MATH3900

Tutorial 3

1. Determine whether each statement is *True* or *False*. Give a short proof for a true statement, or a counterexample with verification for a false statement.
   (a) Let $f(x) = \frac{1}{2}x^T Q x$, where $Q = Q^T \succ 0$. Multiplying f by a positive number leaves its condition number unchanged.
   (b) If the Hessian $H = \nabla^2 f(x)$ is invertible, the Newton direction p, defined by $Hp = -\nabla f(x)$, always minimizes the quadratic Taylor model at x.
   (c) If $H = \nabla^2 f(x) \succ 0$ and $\nabla f(x) \neq 0$, the Newton direction is a descent direction at x.
   (d) Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and convex, and let p be a descent direction at x. If $\alpha > 0$ satisfies the Armijo condition with parameter $c_A \in (0, 1)$, then $\alpha/2$ satisfies the same condition.

Solution: (a) True. Write $\mu = \lambda_{\min}(Q) > 0$ and $L = \lambda_{\max}(Q)$. Let $s > 0$. The Hessian of sf is sQ, so each eigenvalue is multiplied by s. The sharp constants become $s\mu$ and sL. Therefore

$$\kappa(sf) = \frac{sL}{s\mu} = \frac{L}{\mu} = \kappa(f).$$

Both curvatures change, but their ratio does not.

(b) False. Let $f(t) = -t^2/2$ and take $t = 1$. Here $f'(1) = -1$ and $H = f''(1) = -1$, which is invertible. The Newton equation is $-p = 1$, so $p = -1$. However, the quadratic model is

$$m(p) = -\tfrac{1}{2} - p - \tfrac{1}{2}p^2 = -\tfrac{1}{2}(p+1)^2.$$

It has a maximum at $p = -1$, and tends to $-\infty$ as $|p| \to \infty$. Thus it has no minimizer. Invertibility gives a unique stationary point of the model; positive definiteness is what makes that point its unique minimizer.

(c) True. Set $g = \nabla f(x)$. Since $Hp = -g$ and $g \neq 0$, we have $p \neq 0$. Therefore

$$g^T p = (-Hp)^T p = -p^T H p < 0,$$

where symmetry and positive definiteness of H give the last two steps. This is exactly the descent-direction condition. It guarantees decrease for sufficiently small positive steps, not necessarily for a full step.

(d) True. Since $x + \frac{\alpha}{2}p = \frac{1}{2}x + \frac{1}{2}(x + \alpha p)$, the half-step point is a midpoint. Use convexity, then the assumed Armijo inequality:

$$\begin{aligned} f(x + \tfrac{\alpha}{2}p) &\leq \tfrac{1}{2}f(x) + \tfrac{1}{2}f(x + \alpha p) \\ &\leq \tfrac{1}{2}f(x) + \tfrac{1}{2}[f(x) + c_A \alpha \nabla f(x)^T p] \\ &= f(x) + c_A \tfrac{\alpha}{2} \nabla f(x)^T p. \end{aligned}$$

This is Armijo for the half step, with the same c_A. The convexity assumption is essential to this argument.

2. Consider logistic regression with parameter $x = (x_1, x_2)^T$ and six labelled observations:

i	a_i^T	y_i
$1, 2$	$(1, 0)$	$+1$
3	$(1, 0)$	-1
$4, 5$	$(0, 3)$	$+1$
6	$(0, 3)$	-1

$$F(x) = \frac{1}{6}\sum_{i=1}^{6} \log(1 + e^{-y_i a_i^T x}).$$

Define $\sigma(t) = e^t/(1 + e^t)$ and $w(t) = \sigma(t)(1 - \sigma(t))$. You may use

$$\log(1 + e^{-t}) = \log(1 + e^t) - t, \qquad \sigma'(t) = w(t), \qquad 0 < w(t) \le \tfrac{1}{4}.$$

(a) Write $F(x)$ explicitly in terms of x_1, x_2, and simplify it using the identity above. Find $\nabla F(x)$ and $\nabla^2 F(x)$.

(b) Find the smallest global gradient Lipschitz constant L with respect to the Euclidean norm. Explain why a smaller constant cannot work.

(c) Starting from $x^{(0)} = (0, 0)^T$, compute one gradient descent step with step size $1/L$, and one full Newton step. For Newton, solve

$$\nabla^2 F(x^{(0)})p = -\nabla F(x^{(0)}), \qquad x_{\mathrm{N}}^{(1)} = x^{(0)} + p.$$

Compare the two new points with the unique minimizer $x^\star = (\log 2, \frac{1}{3}\log 2)^T$. Which is closer? Does Newton reach $x^\star$ in one step? You may use $\log 2 \approx 0.6931$.

Solution: (a) The first three scores $a_i^T x$ are x_1; the last three are $3x_2$. Each group has two positive labels and one negative label. Therefore

$$F(x) = \tfrac{1}{6}\big[2\log(1 + e^{-x_1}) + \log(1 + e^{x_1}) + 2\log(1 + e^{-3x_2}) + \log(1 + e^{3x_2})\big].$$

For any score t, the given identity gives

$$2\log(1 + e^{-t}) + \log(1 + e^t) = 3\log(1 + e^t) - 2t.$$

Apply it to each group and divide by 6:

$$F(x) = \tfrac{1}{2}\log(1 + e^{x_1}) + \tfrac{1}{2}\log(1 + e^{3x_2}) - \tfrac{1}{3}x_1 - x_2.$$

Since $\frac{d}{dt}\log(1 + e^t) = e^t/(1 + e^t) = \sigma(t)$, the chain rule gives

$$\nabla F(x) = \begin{pmatrix} \frac{1}{2}\sigma(x_1) - \frac{1}{3} \\ \frac{3}{2}\sigma(3x_2) - 1 \end{pmatrix}.$$

Differentiate again. The two terms depend on different coordinates, so the mixed derivatives are zero:

$$\nabla^2 F(x) = \begin{pmatrix} \frac{1}{2}w(x_1) & 0 \\ 0 & \frac{9}{2}w(3x_2) \end{pmatrix}.$$

The factor 3 appears once in the gradient and again when differentiating $\sigma(3x_2)$, giving the factor 9 in the Hessian. Both diagonal entries are positive at every finite x.

Solution: **(b)** The Hessian is diagonal, so its eigenvalues are its diagonal entries. Using $w(t) \le 1/4$,

$$0 < \tfrac{1}{2} w(x_1) \le \tfrac{1}{8}, \qquad 0 < \tfrac{9}{2} w(3x_2) \le \tfrac{9}{8}.$$

Hence $\|\nabla^2 F(x)\|_2 \le 9/8$ for every x. The Hessian characterization of smoothness shows that $L = 9/8$ works.

To check sharpness, note that $\sigma(0) = 1/2$ and $w(0) = 1/4$. Thus

$$H_0 := \nabla^2 F(0) = \begin{pmatrix} 1/8 & 0 \\ 0 & 9/8 \end{pmatrix}.$$

Any global Lipschitz constant must bound the derivative of the gradient at this point. In the direction $e_2 = (0,1)^T$,

$$\lim_{t \to 0} \frac{\|\nabla F(te_2) - \nabla F(0)\|}{|t|} = \|H_0 e_2\| = \tfrac{9}{8}.$$

Therefore no smaller constant works, and the smallest global value is

$$L = \tfrac{9}{8}, \qquad \frac{1}{L} = \tfrac{8}{9}.$$

(c) Substitute $\sigma(0) = 1/2$ into the gradient:

$$g_0 = \nabla F(0) = \begin{pmatrix} 1/4 - 1/3 \\ 3/4 - 1 \end{pmatrix} = \begin{pmatrix} -1/12 \\ -1/4 \end{pmatrix}.$$

The gradient descent step is

$$x_{\mathrm{GD}}^{(1)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \tfrac{8}{9} \begin{pmatrix} -1/12 \\ -1/4 \end{pmatrix} = \begin{pmatrix} 2/27 \\ 2/9 \end{pmatrix}.$$

For Newton, solve the two diagonal equations:

$$\tfrac{1}{8} p_1 = \tfrac{1}{12}, \qquad \tfrac{9}{8} p_2 = \tfrac{1}{4}.$$

Multiplying the first by 8 and the second by 8/9 gives

$$p_1 = \tfrac{2}{3}, \qquad p_2 = \tfrac{2}{9}, \qquad x_{\mathrm{N}}^{(1)} = \begin{pmatrix} 2/3 \\ 2/9 \end{pmatrix}.$$

The points are approximately

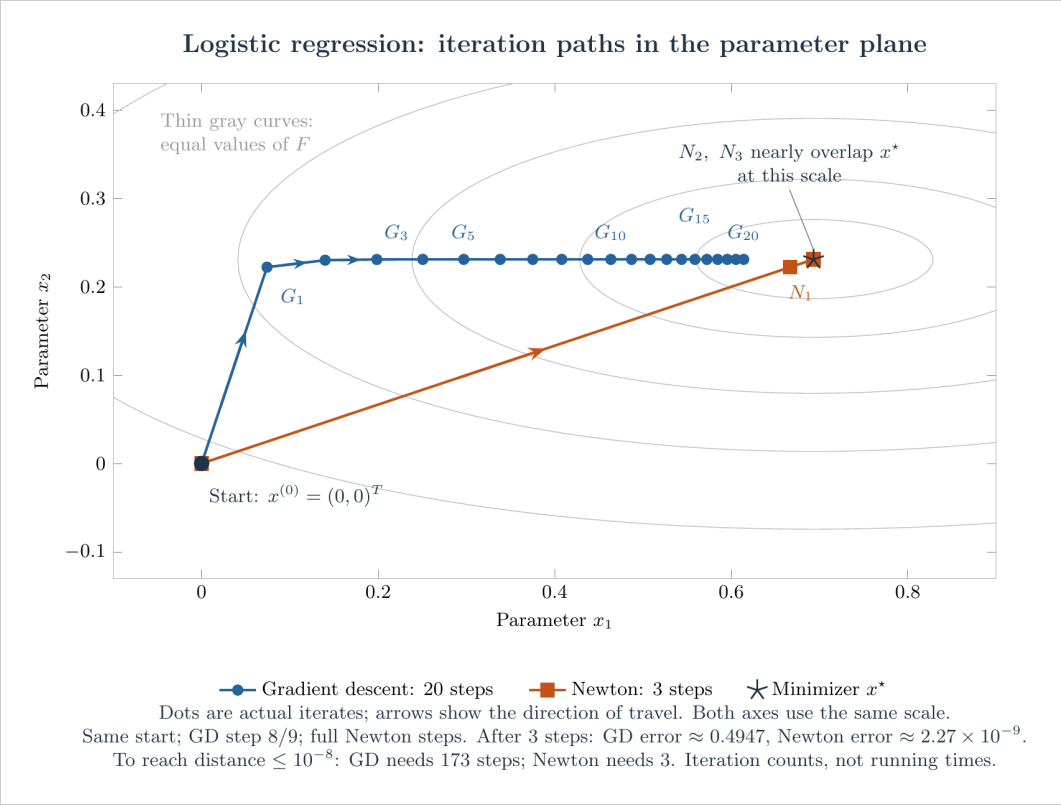
$$\begin{aligned} x^\star &\approx (0.6931, 0.2310)^T, \\ x_{\mathrm{GD}}^{(1)} &\approx (0.0741, 0.2222)^T, \\ x_{\mathrm{N}}^{(1)} &\approx (0.6667, 0.2222)^T. \end{aligned}$$

Both updates have the same second coordinate, but Newton's first coordinate is much closer to $x_1^\star$. Thus Newton is closer in Euclidean distance. Neither point is exactly $x^\star$. Unlike a quadratic objective, this logistic loss has a Hessian that changes with x; its quadratic Taylor model is only a local model.

Solution: (c), continued. To check the supplied minimizer, set both gradient entries to zero. This gives $\sigma(x_1) = \sigma(3x_2) = 2/3$. For either score,

$$\frac{e^t}{1+e^t} = \frac{2}{3} \iff 3e^t = 2 + 2e^t \iff e^t = 2 \iff t = \log 2.$$

This gives $x^\star = (\log 2, \frac{1}{3}\log 2)^T$. Since the Hessian is positive definite everywhere, F is strictly convex, so this stationary point is the unique global minimizer.



Blue points G_k show gradient descent; orange points N_k show Newton. Both start at the origin. Gray curves are level sets of F. After three steps, the distances to $x^\star$ are approximately 0.4947 and 2.27×10^{-9}, respectively. Although N_2, N_3 nearly overlap the star at this scale, neither is exactly the minimizer.

Continuing the same updates, Newton first reaches distance 10^{-8} at iteration 3, while gradient descent first reaches it at iteration 173. The plot shows 20 GD steps and 3 Newton steps; even at G_{20}, the GD distance is still about 0.0791.

The GD step 8/9 is set by the largest possible curvature. At the origin, the first-coordinate curvature is only 1/8, so GD moves slowly in that coordinate. Newton scales each coordinate using its current curvature, giving rapid local convergence here.

These are iteration counts, not running times: Newton needs a Hessian and a linear solve. This example does not guarantee convergence of full Newton steps from every starting point.

3. Let

$$f(t) = \frac{t^3}{3} - t, \qquad t > 0.$$

(a) Compute $f'(t)$ and $f''(t)$, and find the minimizer $t^\star$. At $t^{(0)} = 2$, write the quadratic Taylor model in the displacement p.

(b) Derive the full Newton update for any $t^{(k)} > 0$. Starting from $t^{(0)} = 2$, compute $t^{(1)}$ and $t^{(2)}$.

(c) On $D = [1/2, 3/2]$, show that f'' is 2-Lipschitz and $f''(t) \geq 1$. For $e_k = t^{(k)} - 1$, you may use

$$e_{k+1} = \frac{e_k^2}{2t^{(k)}}.$$

Show that $|e_k| \leq 1/2$ implies $|e_{k+1}| \leq |e_k|^2 \leq 1/4$. Explain how this gives local quadratic convergence.

Solution: **(a)** The derivatives are

$$f'(t) = t^2 - 1, \qquad f''(t) = 2t.$$

On $t > 0$, the derivative is negative before 1 and positive after 1. Thus $t^\star = 1$ is the unique global minimizer on this domain. At $t = 2$, $f(2) = 2/3$, $f'(2) = 3$, and $f''(2) = 4$. Substitute these three values into the quadratic Taylor model:

$$\begin{aligned} m(p) &= f(2) + f'(2)p + \tfrac{1}{2}f''(2)p^2 \\ &= \tfrac{2}{3} + 3p + \tfrac{1}{2}(4)p^2 = \tfrac{2}{3} + 3p + 2p^2. \end{aligned}$$

(b) The Newton equation is $2t^{(k)}p^{(k)} = -((t^{(k)})^2 - 1)$. Since $t^{(k)} > 0$, we may divide by $2t^{(k)}$, giving

$$\begin{aligned} t^{(k+1)} &= t^{(k)} - \frac{(t^{(k)})^2 - 1}{2t^{(k)}} \\ &= \frac{(t^{(k)})^2 + 1}{2t^{(k)}} = \tfrac{1}{2}\left(t^{(k)} + \frac{1}{t^{(k)}}\right). \end{aligned}$$

In particular,

$$t^{(1)} = \tfrac{1}{2}\left(2 + \tfrac{1}{2}\right) = \tfrac{5}{4}, \qquad t^{(2)} = \tfrac{1}{2}\left(\tfrac{5}{4} + \tfrac{4}{5}\right) = \tfrac{1}{2} \cdot \tfrac{41}{20} = \tfrac{41}{40}.$$

Solution: **(c)** For $u, v \in D$,

$$|f''(u) - f''(v)| = |2u - 2v| = 2|u - v|.$$

Thus $\rho = 2$ is a Hessian Lipschitz constant. Also $f''(t) = 2t \geq 1$ on D. Subtract 1 from the Newton formula:

$$\begin{aligned} e_{k+1} &= \frac{(t^{(k)})^2 + 1 - 2t^{(k)}}{2t^{(k)}} \\ &= \frac{(t^{(k)} - 1)^2}{2t^{(k)}} = \frac{e_k^2}{2t^{(k)}}. \end{aligned}$$

If $|e_k| \leq 1/2$, then $t^{(k)} \in D$, so $2t^{(k)} \geq 1$. It follows that

$$0 \leq e_{k+1} \leq e_k^2 \leq \tfrac{1}{4} \leq \tfrac{1}{2}.$$

Starting with $|e_0| \leq 1/2$, induction shows that all iterates remain in D and the same estimate applies at every step. In this interval, $e_k^2 \leq |e_k|/2$, so the errors tend to zero. The stronger estimate $|e_{k+1}| \leq |e_k|^2$ shows that the convergence is quadratic. The stated estimate is local because it was established for starts within $1/2$ of the solution.

4. Let

$$f(x) = \tfrac{1}{2}(x_1^2 + 3x_2^2), \qquad x = (3, 1)^T, \qquad g = \nabla f(x), \qquad p = -g.$$

(a) Compute g, $g^T p$, and $\phi(\alpha) = f(x + \alpha p)$. Find the exact line-search step over $\alpha \geq 0$, the new point, and its objective value.

(b) With $c_A = 1/4$, solve the Armijo inequality

$$f(x + \alpha p) \leq f(x) + c_A \alpha g^T p$$

to find all acceptable positive step sizes.

(c) Check $\alpha = 1/4$ and $\alpha = 5/6$. For each, give the objective value and the Armijo bound. Use these two trials to distinguish an exact line-search step, a step satisfying Armijo, and a step that merely decreases f.

Solution: **(a)** The gradient is $\nabla f(x) = (x_1, 3x_2)^T$. At the given point,

$$f(x) = \tfrac{1}{2}(9 + 3) = 6, \quad g = (3, 3)^T, \quad p = (-3, -3)^T, \quad g^T p = -18.$$

Along this direction, $x + \alpha p = (3 - 3\alpha, 1 - 3\alpha)^T$, so

$$\begin{aligned} \phi(\alpha) &= \tfrac{1}{2}\left[(3 - 3\alpha)^2 + 3(1 - 3\alpha)^2\right] \\ &= \tfrac{1}{2}\left[9 - 18\alpha + 9\alpha^2 + 3 - 18\alpha + 27\alpha^2\right] \\ &= 6 - 18\alpha + 18\alpha^2. \end{aligned}$$

Differentiate with respect to α:

$$\phi'(\alpha) = -18 + 36\alpha, \qquad \phi''(\alpha) = 36 > 0.$$

Thus the unique exact step is $\alpha_\star = 1/2$, which is positive. It gives

$$x_{\text{exact}} = (3,1)^T + \tfrac{1}{2}(-3,-3)^T = (3/2, -1/2)^T,$$

$$f(x_{\text{exact}}) = \tfrac{1}{2}\left(\tfrac{9}{4} + 3 \cdot \tfrac{1}{4}\right) = \tfrac{3}{2}.$$

This also agrees with the quadratic formula $\alpha_\star = (g^T g)/(g^T Q g) = 18/36$, where $Q = \text{diag}(1,3)$.

(b) The Armijo bound is

$$f(x) + c_A \alpha g^T p = 6 + \tfrac{1}{4}\alpha(-18) = 6 - \tfrac{9}{2}\alpha.$$

Substitute $\phi(\alpha)$ and rearrange:

$$\begin{aligned} 6 - 18\alpha + 18\alpha^2 &\le 6 - \tfrac{9}{2}\alpha \\ 18\alpha^2 &\le \tfrac{27}{2}\alpha. \end{aligned}$$

Since $\alpha > 0$, division by 18α gives

$$0 < \alpha \le \tfrac{3}{4}.$$

This is an interval of acceptable steps, not a rule selecting one particular step.

(c) For $\alpha = 1/4$,

$$\phi(1/4) = 6 - \tfrac{18}{4} + \tfrac{18}{16} = \tfrac{21}{8}, \qquad 6 - \tfrac{9}{2} \cdot \tfrac{1}{4} = \tfrac{39}{8}.$$

Since $21/8 \le 39/8$, Armijo holds. But $21/8 > 3/2$, so this trial is not the exact line-search step.

For $\alpha = 5/6$,

$$\phi(5/6) = 6 - 15 + \tfrac{25}{2} = \tfrac{7}{2}, \qquad 6 - \tfrac{9}{2} \cdot \tfrac{5}{6} = \tfrac{9}{4}.$$

Here $7/2 < 6 = f(x)$, so the objective decreases. However, $7/2 > 9/4$, so the trial fails Armijo.

In this example, the exact step $1/2$ is also Armijo-acceptable. The trial $1/4$ is acceptable but not exact, and the trial $5/6$ decreases the objective without giving the decrease required by Armijo.